# Xiomara T Gonzalez

*email* xiomarag@utexas.edu
*website* https://sites.utexas.edu/xiomara/

## Research Interests

information-seeking behavior, natural language processing, image and video processing, qualitative and quantitative research methods, clinical decision-making, accessibility, health equity, engineering education

## Education

**University of Texas at Austin** **Expected Summer 2027**
Ph.D. in Electrical and Computer Engineering (ECE)
*Advisor: Mia K. Markey*
Graduate Certificate in Engineering Education

**University of Texas at Austin** **2021 - 2023**
M.S.E. in Electrical and Computer Engineering (ECE)

**University of California at Berkeley** **2015 - 2019**
B.S. in Electrical Engineering and Computer Science (EECS)

## Research Experience

**Graduate Research Fellow** **06/2021 - present**
*Biomedical Informatics Lab - Austin, TX*

- Research focuses on leveraging machine learning and multimedia (text, image, video) processing techniques to develop scalable health information tools that help users synthesize health information collected online
- Conducting qualitative and quantitative (mixed-methods) studies to understand user attitudes and perceptions toward online and AI-generated health information
- Concurrently holding a Graduate Trainee Appointment at MD Anderson Cancer Center's Department of Plastic Surgery (MBRRP)

**Graduate Research Assistant** **01/2024 - 12/2024**
*Center for Equity in Engineering (UT) - Austin, TX*

- Research focused on leveraging open-source large language models (e.g., Llama3) to streamline and automate tasks part of qualitative data analysis pipelines

**Bioinformatics and Data Science Intern** **06/2023 - 12/2023**
*Washington University School of Medicine - St. Louis, MO*

- Conducted a mixed-methods study focused on surveying the general public and surgical patients' attitudes toward the use of AI and ML technologies in perioperative care (before, during, and after surgery)

**micro-CT Research Intern** **06/2018 - 12/2018, 06/2019 - 08/2019**
*Advanced Light Source - Lawrence Berkeley National Laboratory - Berkeley, CA*

- Automated fiber center detection in ceramic-matrix composites (gas turbine material) datasets using a mixed-scale dense deep convolutional neural network (MS-D DCNN) implemented in Python/PyTorch
- Became familiar with micro-computed tomography instrumentation and data acquisition techniques at beamline 8.3.2

## Teaching Experience

**Graduate Teaching Assistant** **09/2023 - 12/2023, 09/2025 - 12/2025**
*Biomedical Engineering (BME) Department - UT Austin*

- Led three laboratory sections (∼20 students each) focused on programming in C++ and Python for an 85-student undergraduate course (BME 303: Introduction to Computing)
- Developed interactive and engaging material for lab sections (e.g., gamification of review problems)
- Assisted with clerical tasks relevant for course instruction

## Industry Experience

**Embedded Software Engineer Intern** **01/2020 - 08/2020**
*Schlumberger Technology Co. - Sugar Land, TX*

- Developed an algorithm to automate the calibration of two piston pumps part of Ora, an intelligent wireline formation testing platform
- Leveraged Microsoft Azure IoT services to create a pipeline that enabled remote management and control of downhole assets
- Reworked an existing code base for a tool simulator to add compatibility with CAN bus communications and Azure IoT SDKs
- For proof of concept, remote firmware upgrades on the simulator's data acquisition unit, power supply controller, & FPGAs were performed

## Refereed Journal Publications

[1] Wang H, **Gonzalez XT**, Renta Lopez GA, Bordes MC, Hout MC, Choi SW, Reece GP, Markey MK. Breast cancer survivors' perceptual map of breast reconstruction appearance outcomes. *Journal of Medical Imaging* 2025;12(5):051802. doi:10.1117/1.JMI.12.5.051802

[2] **Gonzalez XT**, Roubaud MS, Schaverien MV, Largo RD, Parham CS, Francis AM, Chen TA, Hoffman A, Dickey RM, Markey MK, Reece GP. Assessment of appearance-related questions about breast reconstruction generated by ChatGPT.*Plastic and Reconstructive Surgery - Global Open* 2025;13(3):e6625. doi:10.1097/GOX.0000000000006625

[3] **Gonzalez XT**, Steger-May K, Abraham J. Just another tool in their repertoire: Uncovering insights into public and patient perspectives on clinicians' use of machine learning in perioperative care. *Journal of the American Medical Informatics Association.* 2025;32(1):150-162. doi:10.1093/jamia/ocae257

[4] Mehta D, **Gonzalez XT**, Huang G, Abraham J. Machine learning-augmented interventions in perioperative care: a systematic review and meta-analysis. *British Journal of Anaesthesia.* 2024;133(6):1159-1172. doi:10.1016/j.bja.2024.08.007

[5] Nicklaus KM, **Gonzalez XT**, Sachdev K, Liu J, Chopra D, Hoffman AS, Hanson SE, Markey MK, Reece GP. What Does "Dr. Google" Show Patients Searching for Breast Reconstruction Outcomes Photographs? *Plastic and Reconstructive Surgery - Global Open.* 2022;10(5):e4331. doi:10.1097/GOX.0000000000004331

## under review

[1] **Gonzalez XT**, Denton M, Deters J, Coloyan Flemming G, Katz A. Sentiment analysis of engineering study abroad experiences: A behavioral cultural intelligence perspective using large language models. (under review)

[2] **Gonzalez XT**, Dickey RM, Upshaw SU, Markey MK. Beyond Mobile Phones: The Role of Smart TVs in FemTech. (under review)

## Poster Presentations

[1] **Gonzalez XT**, Roubaud MS, Schaverien MV, Largo RD, Parham CS, Francis AF, Chen TA, Hoffman A, Markey MK, Reece GP, "Appearance-related questions about breast reconstruction generated by ChatGPT: Potential to contribute to informed consent and shared decision making," Biomedical Engineering Society Annual Meeting. Baltimore, MD. October 2024.

[2] **Gonzalez XT**, Roubaud MS, Schaverien MV, Largo RD, Parham CS, Francis AF, Chen TA, Hoffman A, Markey MK, Reece GP, "Dear ChatGPT: How Do I Ask My Doctor What I Will Look Like After Breast Reconstruction?," AI in Health Conference. Houston, TX. September 2024.

[3] **Gonzalez XT**, Meng A, Jamro E, Abraham J, "General Public and Patient Perspectives and Attitudes towards Use of Machine Learning in Perioperative Care," Institute for Informatics, Data Science, and Bioinformatics. St. Louis, MO. August 2023.

[4] Nicklaus K, **Gonzalez XT**, Sachdev K, Liu J, Chopra D, Hoffman AS, Hanson SE, Markey MK, Reece GP, "Beyond Bookmarks and Binders: Deciphering Dr. Google to Enhance Patient-Provider Communication," NIH-NIBIB Training Grantees Meeting. Bethesda, MD. June 2023.

[5] **Gonzalez XT**, MacNeil JML, Ushizima DM, Barnard HS, Parkinson DY, "Automating Fiber Detection in Ceramic-Matrix Composites Using a Mixed-Scale Dense Deep Convolutional Neural Network," UC Berkeley LSAMP Summer Research Symposium. Berkeley, CA. August 2019.

[5] **Gonzalez XT**, Pelt D, Sethian J, Perciano T, Ushizima D, Barnard H, Parkinson D, "Three-dimensional Visualizations of Ceramic-Matrix Composites in Controlled Environments," Society of Hispanic Professional Engineers (SHPE) National Conference - Engineering Science Symposium. Cleveland, OH. November 2018.

## Talks

**Assessment of appearance-related questions about breast reconstruction generated by ChatGPT** 10/2025
The University of Texas at Austin Cancer Research Retreat - Quantitative Oncology and Research Applications

**Beyond Bookmarks and Binders: Deciphering Dr. Google to Enhance Patient-Provider Communication** 03/2023
Computing Research Association (CRA) Grad Cohort Workshop for IDEALS

## Honors & Awards

| | |
|---|---|
| UT Tolbert Fellowship in Electrical and Computer Engineering | 2025 - 2026 |
| Cadence Design Systems Diversity in Technology Scholarship | 2023 |
| UT Cockrell School of Engineering DEI Seed Grant (Co-Lead) | 2023 |
| NIH Ruth L. Kirschstein National Research Service Award | 2022 - 2023 |
| SHPE Helen Cuesta Graduate Scholarship | 2022 |
| UT Thrust 2000 - W.H. Cunningham Endowed Graduate Fellowship in Engineering | 2021 - 2025 |
| UT Tolbert Fellowship in Electrical and Computer Engineering | 2021 |
| UC Berkeley NSF Louis Stokes Alliance for Minority Participation Scholar | 2019 |
| UC Berkeley CS Scholar | 2015 - 2016 |
| UC Berkeley College of Engineering Pre-Engineering Program Scholar | 2015 |
| UC Berkeley College of Engineering Ford Scholarship | 2015 |
| National Center for Women and Information Technology AiC Award - SF Bay Area | 2015 |
| Johns Hopkins Center for Talented Youth Goldman Sachs Scholar | 2011 |

## Professional Activities

**External Reviewer**
Journal of the American Medical Informatics Association
Journal of the American Medical Informatics Association (JAMIA) Open
Medical Decision Making (MDM) Policy and Practice

## Affiliations

**Graduate Student Mentor** 2023 - 2025
*Society for Advancing Gender Equity in STEM (SAGES)*
- Have served as mentor for four cycles, Spring 2023 - Fall 2024

**Public Relations Officer, Annual Retreat Committee** 2022 - 2024
*Graduates for Underrepresented Minorities (GUM) at UT Austin*
- Coordinate and promote seminars and workshops for the professional development of GUM members
- Contributions led to GUM being awarded the 2024 Best Social Organization Award from UT Austin's Dean of Students

**Alumni Application Reviewer** 2016 - present
*National Center for Women and Information Technology (NCWiT)*
- Aspirations in Computing Award application reviewer for the high school level awards

**Vice President, Webmaster** 2017-18, 2018-19
*Society of Hispanic Professional Engineers (SHPE, UC Berkeley Chapter)*

- Advised the administrative board on highly attended events (e.g., Graduate School Workshop Series, K-12 outreach events)
- Helped secure corporate sponsorships up to $20,000/year to fund SHPE events (e.g., Winter Academic Training Camp)
- Led department town halls and collaborated with the Associate Dean of Equity and Inclusion to revise policy mandating diversity training for faculty and student instructors
- Contributions led to SHPE being awarded the 2019 Most Outstanding Student Organization Award from Berkeley's Dean of Student Affairs

**Certifications**

CITI Program - Good Clinical Practice - Clinical Trials with Investigational Drugs and Medical Devices
CITI Program - Human Research - Biomedical Researchers
CITI Program - Responsible Conduct of Research for Engineers
UT Austin Inclusive Classrooms Leadership Certificate
UT Austin Center for Teaching and Learning Teaching Preparation Certificate

**Memberships**

American Society for Engineering Education (ASEE)
Association for Computing Machinery (ACM)
Biomedical Engineering Society (BMES)
Institute of Electrical and Electronics Engineers (IEEE)
Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
Society of Professional Hispanic Engineers (SHPE)

**Coursework**

**UT Austin:** Natural Language Processing and Applications (INF385T), Mobile Computing (ECE 382V), Digital Video Processing (ECE 381K), Human Signals: Sensing and Analytics (ECE 382V), Spoken Language Technologies (ECE 381V), Data Mining (ECE 380L), Machine Learning (INF 385T), Computational Imaging/Inverse Problems (CSE 397), Biomedical Imaging Modalities (ECE 385J-18), Imaging Clinical Immersion (BME 385J), Functional Imaging Lab (BME 381J-8), Qualitative Research Methods in Engineering (ME 397)

**UT Austin - Engineering Education Certificate:** Curriculum Design and Assessment in Engineering (ME 398S), Interdisciplinary, Transdisciplinary, and Hybrid Learning (STM f390T), Teaching Engineering (ME 398T), Engineering Teaching Practicum (ME 698C)

**UC Berkeley:** Medical Imaging (EECS C145B), Signals and Systems (EE 120), Artificial Intelligence (CS 188), Robotics (EECS 106), Operating Systems (CS 162), 3D Modeling and Design, Modern Physics

* in progress; UT = UT Austin; updated: March 2026